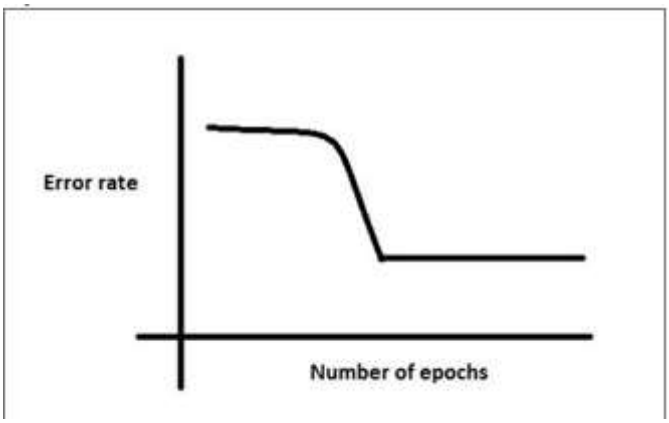
You completed this test on *18/02/2022, 15:00*
Your score is 74.07%

In training a neural network, you notice that the loss does not decrease in the few starting epochs.



The reasons for this could be:

As the problem can be solved using all the techniques

1. The learning is rate is low
2. Regularization parameter is high
3. Stuck at local minima

What according to you are the probable reasons?

1 and 2

2 and 3

1 and 3

Any of these

Which of these are popular regularization techniques? (Select all that apply)

Dropout

Gradient Clipping is not so popular regularization technique

L1

L2

Gradient Clipping

What is true of L1, L2 and Dropout regularization methods? (Select all that apply)

L1, L2 regularization methods modify the loss function whereas Dropout modifies the ANN itself

Dropout modifies the loss function whereas L1, L2 regularization modifies the ANN itself

Both Dropout and L1, L2 regularization methods modify the loss function

Both Dropout and L1, L2 regularization methods modify the ANN itself

L1 and L2 is applied to reduce or decrease Loss or Cost function whereas Dropout is used to switch off the some percent of neurons in layer randomly.

Y = ax^2 + bx + c (polynomial equation of degree 2)
Can this equation be represented by a neural network of single hidden layer with linear threshold?

Yes

A polynomial equation cannot be achieved by linear threshold

No

What are the disadvantages of Sigmoid activation functon? (Select All that apply)

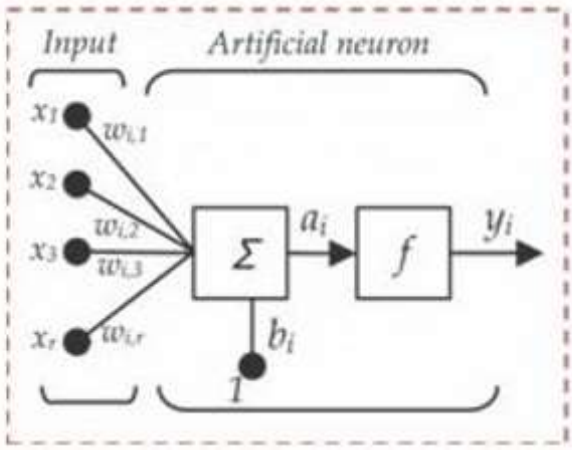
Sigmoid is a Non zero Centered Activation Function

Models converge slowly when using Sigmoid

Sigmoid is a linear activation function

Vanishing Gradient Problem

Below is a mathematical representation of a neuron.



The different components of the neurons are denoted as:

1. **x1, x2,…, xN: These are inputs to the neuron. These can either be the actual observations from input layer or an intermediate value from one of the hidden layers.**
2. **2w1, w2,…,wN: The Weight of each input.**
3. **bi: Is termed as Bias units. These are constant values added to the input of the activation function corresponding to each weight. It works similar to an intercept term.**
4. **a: Is termed as the activation of the neuron**
5. **y: is the output of the neuron**

$$a = f(\sum_{i=0}^{N} w_i x_i)$$

Considering the above notations, will a line equation (y = mx + c) fall into the category of a neuron?

Yes

Yes linear function can also be represented as neural network

No

What are the advantages of ReLU activation function? (Select All that apply)

- *ReLU solves the vanishing gradient problem.*
- *ReLU allows for faster convergence*
- ReLU filters out negative values
- ReLU always gives better validation accuracy

ReLU = max(z,0)

so it would filter negative values

it also solves vanishing gradients

it also helps in faster convergence

but it does not guarantee to give better validation accuracy

Which activation function and loss function would you use for a binary classification problem?

- Softmax with MSE Loss
- TanH with KLD Loss
- Linear Activation with Hinge Loss
- *Sigmoid with Binary Crossentropy*

MSE is for regression not classification

Linear activation cannot be used for classification

What are the disadvantages of ReLU activation function? (Select All that apply)

- *Dead Neurons*
- High computational complexity
- Makes an ANN more susceptible to overfitting
- All of these

As negative values will be converged as zero it creates dead neurons

ReLU = max(0,z)

It does not have high computation

It reduces overfitting

Consider the scenario. The problem you are trying to solve has a small amount of data. Fortunately, you have a pre-trained neural network that was trained on a similar problem. Which of the following methodologies would you choose to make use of this pre-trained network?

- Re-train the model for the new dataset
- Assess on every layer how the model performs and only select a few of them
- Fine tune the last couple of layers only
- *Freeze all the layers except the last, re-train the last layer*

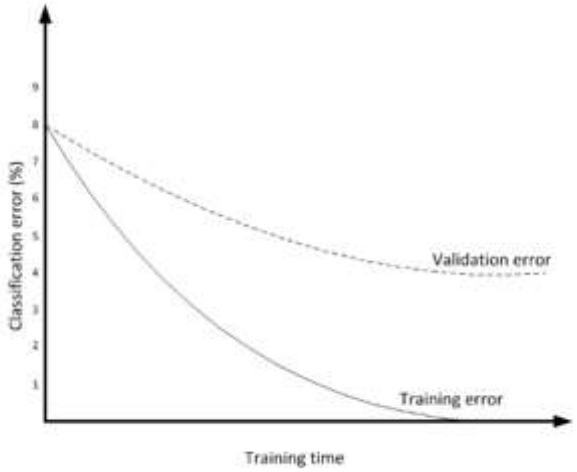
As we have pretrained model retraining would add more computation time

Assessing on every layer is time consuming

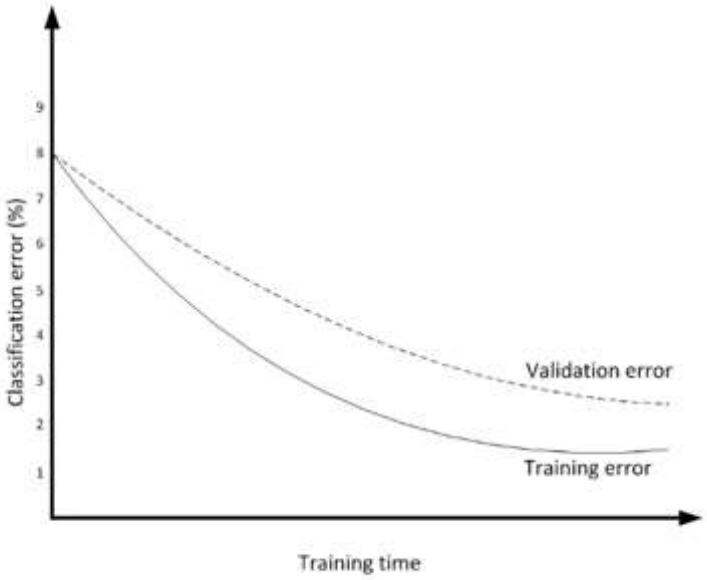
This may be a better idea as the data is small we can use other approach

You are experimenting with two different models for a classification task. The figures below show the classification error you get as training progresses on the training data and the validation data for each of the two models. Which model do you think would perform better on previously unseen test data?

1.



2.



Only 1

Only 2

Observing both the images we can say that the second model does perform better as the first model is suspected to overfitting

Both

An artificial neuron receives n inputs x1, x2, x3.....xn with weights w1, w2, ...wn attached to the input links. The weighted sum_____ is computed to be passed on to a non-linear filter Φ called activation function to release the output.

Σ wi

Σ xi

Σ wi + Σ xi

Σ wi xi*

What is delta (error) in the perceptron model of a neuron?

error due to environmental condition

difference between desired & target output E = (y - ycap)

can be both due to the difference in target output or environmental condition

none of the mentioned

Suppose you have a multi-class classification problem with three classes, trained with a 3 layer network. Let a(3)1=(hΘ(x))1 be the activation of the first output unit, and similarly a(3)2=(hΘ(x))2 and a(3)3=(hΘ(x))3. Then for any input x, it must be the case that a(3)1+a(3)2+a(3)3=1. (True/False) [Explain your choice]

False , The outputs of the units may not be equal to 1

True.
As all the output layers output should be equal to 1.
say for example there are 3 classes namely 0 1 and 2, and their probabilities should be 1, like ((1:0.5),(2:0.3),(3:0.2)), the model is likely to predict that instance as output 1 as it has a higher probability

---

What is the effect of L2 regularization? (Select all that apply)

Promotes model sparsity

Reduces model expressiveness

It does all the activities that are mentioned.
It promotes model sparsity, it reduces model expressiveness as it decays some weights

Drives model weights close to zero but not completely zero

All of these

What is the effect of L1 regularization? (Select all that apply)

Promotes model sparsity

Reduces model expressiveness

It does all the activities that are mentioned.
It promotes model sparsity, it reduces model expressiveness as it decays some weights

Drives model weights to zero

All of these

Which neural network model is computationally expensive

Back Propagation

LVQ

Back propagation is computationally expensive considering all the mentioned models

ART

Recurrent Back Propagation

A neuron with 3 inputs has the weight vector [0.2 -0.1 0.1]^T and a bias θ = 0. If the input vector is X = [0.2 0.4 0.2]^T then the total input to the neuron is:

0.2

weight vector (w) = [0.2 -0.1 0.1]T
input vector (xT) = [0.2 0.4 0.2]T
bias (b) = 0
wT*X + b

10.0

0.02

i.e. = [0.2 –0.1 0.1] * [0.2 0.4 0.2]T + 0
= [0.2 * 0.2 – 0.1 * 0.4 + 0.1 * 0.2] + 0
= 0.02

-1

A 4-input neuron has weights 1, 2, 3 and 4. The transfer function is linear with the constant of proportionality being equal to 2. The inputs are 4, 10, 5 and 20 respectively. The output will be:

238

2*(1*4 + 2*10 + 3*5 + 4*20) = 238

76

119

A 4-input neuron has weights 1, 2, 3 and 4. The transfer function is linear with the constant of proportionality being equal to 2. The inputs are 4, 10, 5 and 20 respectively. The output will be:

238

2*(1*4 + 2*10 + 3*5 + 4*20) = 238

76

119

Batch Normalization is helpful because

It is used to normalize the inputs so that dataset will be balanced.

It normalizes (changes) all the input before sending it to the next layer

It returns back the normalized mean and standard deviation of weights

It is a very efficient backpropagation technique

None of these

The activation values of the hidden units in a neural network, with the sigmoid activation function applied at every layer, are always in the range (0, 1). (True/False)

True Sigmoid activation function lie in the range 0, 1

False

What activation function does this equation represent? (e^x – e^-x) / (e^x + e^-x)

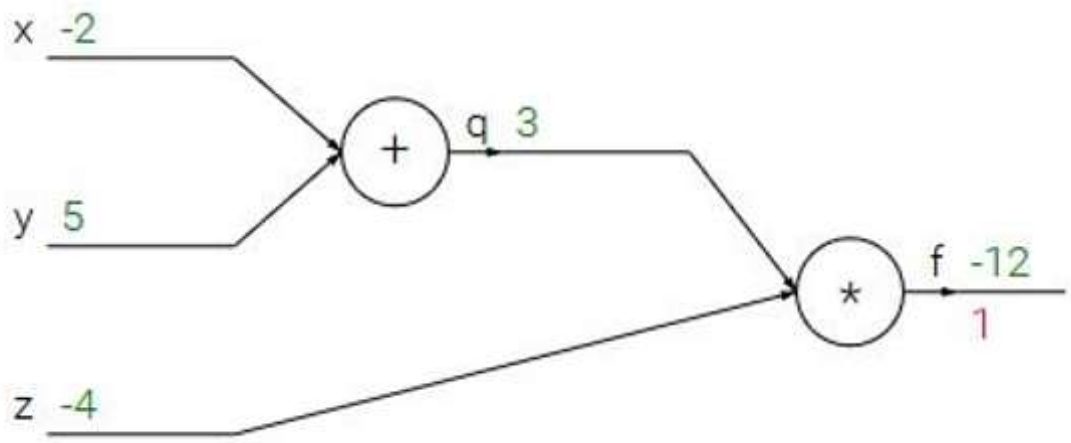
Softmax

Tanh

Sigmoid

ReLU

Suppose you have inputs as x, y, and z with values -2, 5, and -4 respectively. You have a neuron 'q' and neuron 'f' with functions:
q = x + y
f = q * z
Graphical representation of the functions is as follows:



What is the gradient of F with respect to x, y, and z?
(HINT: To calculate gradient, you must find (df/dx), (df/dy) and (df/dz))

(-3,4,4)

(4,4,3)

(-4,-4,3)

(3,-4,-4)

Identify the following activation function :

φ(V) = Z + (1/ 1 + exp (– x * V + Y)), Z, X, Y are parameters

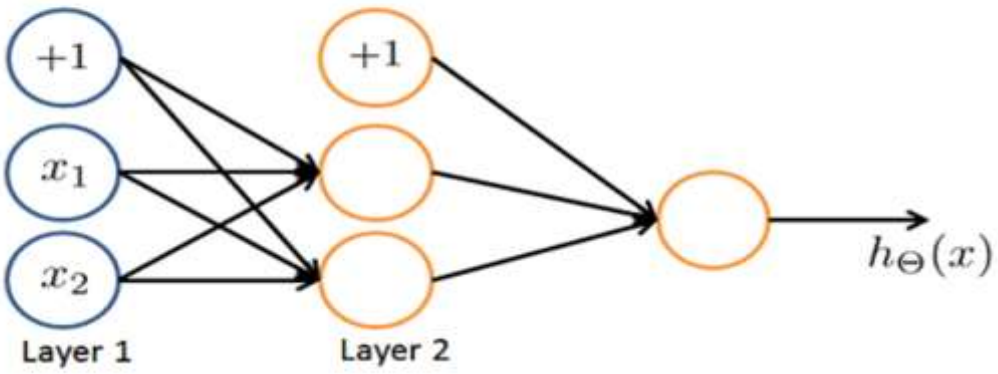
Step Function

Ramp Function

Sigmoid function

Gaussian Function

You are using the neural network pictured below and have learned the parameters $\Theta^{(1)} = \begin{bmatrix} 1 & 1 & 2.4 \\ 1 & 1.7 & 3.2 \end{bmatrix}$ (used to compute $a^{(2)}$) and $\Theta^{(2)} = \begin{bmatrix} 1 & 0.3 & -1.2 \end{bmatrix}$ (used to compute $a^{(3)}$) as a function of $a^{(2)}$). Suppose you swap the parameters for the first hidden layer between its two units so $\Theta^{(1)} = \begin{bmatrix} 1 & 1.7 & 3.2 \\ 1 & 1 & 2.4 \end{bmatrix}$ and also swap the output layer so $\Theta^{(2)} = \begin{bmatrix} 1 & -1.2 & 0.3 \end{bmatrix}$. How will this change the value of the output $h_\Theta(x)$?



It will stay the same

It will increase

It will decrease

Insufficient information to tell, it may increase or decrease

a1 = activation(1+ x1 * 1 + x2 * 2.4)
b1 = activation(1+ x1 * 1.7 + x2 * 3.2)

a2 = activation(1+ x1 * 1.7 + x2 * 3.2)
b1 = activation(1+ x1 * 1 + x2 * 2.4)

1 + 0.3 * a1 – 1.2 * b1
1+(-1.2)*b1 + 0.3 * a1

Both equations are same

What is the effect of Dropout? (Select all that apply)

Promotes model sparsity

Reduces model expressiveness

Makes an ANN robust by reducing dependence on specific neurons

All of these

Dropout switches off the neurons randomly which reduces model expressiveness towards single output and it also makes model as sparse.
Dropout induces self-learning which reduces dependence on other neurons.

Passed. Let's continue.